Chunyu Jiang

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EDUCATION

University of Pennsylvania | *Philadelphia, PA* Aug 2025 – May 2027 (expected)
Master of Science in Engineering in *Robotics* **GPA:** 3.77/4.00
Tongji University | *Shanghai, China* Sep 2021 – Jun 2025
Bachelor of Engineering in *Mechatronics Engineering* **GPA:** 90.34/100 (Rank 2)

PUBLICATIONS

** equal contribution*

Residual Compliant Policy: Learning Compliant Policies from Few Demonstrations
Chunyu Jiang*, Yichen Cai*, Shuai Li, Wenbin Sun, João Carvalho, Kostas Daniilidis, Oleg Arenz, Cristiana de Farias, Jan Peters
Under review at ICRA 2027 [Project Page]
Learns compliant contact-rich manipulation from 7 teleoperated demonstrations via simulation augmentation (Laplacian editing + trajectory optimization), a flow-matching policy commanding an admittance controller, and residual RL. 40/40 out-of-distribution real-robot successes vs. 33/40 for an adaptive-compliance baseline.

HandMimic: Learning Dexterous Manipulations from Videos
Zhuoyang Pan, Yunzhou Song, **Chunyu Jiang**, Junyao Shi, Jiahui Lei, Kostas Daniilidis
Under review at ICLR 2027
Learns dexterous manipulation from monocular human videos by treating the reconstructed hand–object interaction as an imperfect reference for RL in simulation, practiced across generated "digital siblings" of the object. 85.7% success vs. 35.4% for ManipTrans from reconstructed demonstrations; 56/60 real-world replays on a KUKA iiwa 7 + XHand.

RESEARCH EXPERIENCE

GRASP Lab, University of Pennsylvania | *Philadelphia, PA* Oct 2025 – present
*Research Assistant, advised by **Prof. Kostas Daniilidis** and **Prof. Nadia Figueroa***

- Built a vision-based arm–hand teleoperation and multimodal data-collection system for a KUKA iiwa 7 with a tactile XHand (iPhone VIO for the arm, camera-based AnyTeleop for the hand).
- Contributed to HandMimic (under review at ICLR 2027) on learning dexterous manipulation from human videos.
- **Master's thesis (ongoing):** compliant policies for forceful tool use with a dexterous hand, where both the arm (impedance control) and the hand (fingertip tactile sensing) regulate contact force.

Intelligent Autonomous Systems Group, TU Darmstadt | *Darmstadt, Germany* May 2026 – Sep 2026
*Visiting Student Researcher, hosted by **Prof. Jan Peters***

- Co-first author of Residual Compliant Policy (under review at ICRA 2027): built the admittance-control execution stack, the simulation data-augmentation pipeline, the flow-matching compliant base policy, and the residual RL policy trained in Newton.
- Evaluated over 520 real-robot trials on a Franka FR3: 40/40 out-of-distribution successes vs. 33/40 for Adaptive Compliance Policy and 24/40 for Laplacian-editing augmentation.

PROJECTS

Autonomous Drone Racing with Reinforcement Learning | *UPenn* 2026
Trained a quadrotor policy with RL in Isaac Lab to race through a track at high speed, then deployed it on a real drone.
Low-Cost Teleoperation System for Robot Manipulation | *Tongji University* 2024
Built a low-cost teleoperation system (sensorized 3D-printed leader arm, KUKA follower), collected RGB and proprioceptive data, and trained an ACT policy that grasps and assembles workpieces on the real robot.

INDUSTRY EXPERIENCE

Volkswagen Commercial Vehicles, Volkswagen Group | *Wolfsburg, Germany* Mar 2025 – Jun 2025
Intern, NE-GI/T

HONORS & AWARDS

Outstanding Graduate of Tongji University (Top 5%), 2025
Outstanding Student of Tongji University (Top 5%), 2023
Tongji University First-Class Scholarship (Top 10%), 2023
DAAD Scholarship (Ranked 1st), 2024
Klaus Fischer Scholarship (Top 10%), 2024

SKILLS

Programming: Python, C++, C, MATLAB
Learning & Simulation: PyTorch, ROS 2, Isaac Lab, Newton, MuJoCo
Hardware & Control: Franka FR3, KUKA iiwa 7, XHand, DexUMI, Haption Virtuose 6D, F/T sensors, admittance/impedance control, 3D printing
Languages: Chinese (native), English (proficient), German (proficient, TestDaF 20/20)